

# Ritabrata Chakraborty

[GitHub](#) | [Google Scholar](#) | [Portfolio](#) | [LinkedIn](#) | [ritabratbits@gmail.com](mailto:ritabratbits@gmail.com) | +91 89107 83548

## EDUCATION

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| <b>University of Pennsylvania</b><br><i>M.S.E. in Robotics, GPA: -/4</i>                                                                                                                                                                                                                        | <b>Pennsylvania, USA</b><br><i>Aug '26 – May '28</i> |
| <b>Birla Institute of Technology and Science, Pilani (BITS Pilani)</b><br><i>B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10</i> <ul style="list-style-type: none"><li><b>Institute Merit-cum-Need Scholarship</b> — Consecutive Recipient (Top 8% of Students)</li></ul> | <b>Rajasthan, India</b><br><i>Oct '22 – May '26</i>  |

## LEADERSHIP & VOLUNTEERING

|                                                                                                                                                                                                                                                                                                                                                                                                              |                          |
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| <b>Deputy Director of Academic Programming</b><br><i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>                                                                                                                                                                                                                                                                      | <i>Aug '26 – Present</i> |
| <b>Vice-President</b><br><i>South Asian Association (Rangoli), University of Pennsylvania</i>                                                                                                                                                                                                                                                                                                                | <i>Aug '26 – Present</i> |
| <b>President &amp; Secretary</b><br><i>Mechanical Engineering Association (MEA), BITS Pilani</i> <ul style="list-style-type: none"><li>Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction.</li><li>Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.</li></ul>                         | <i>Jun '24 – May '26</i> |
| <b>President &amp; Tech Fest Coordinator</b><br><i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i> <ul style="list-style-type: none"><li>Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts.</li><li>Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.</li></ul> | <i>Oct '24 – Jul '25</i> |
| <b>Project Manager</b><br><i>Tinkerer's Lab (TL), BITS Pilani</i> <ul style="list-style-type: none"><li>Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects.</li><li>Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.</li></ul>                                                                           | <i>May '24 – Jul '25</i> |
| <b>Treasurer</b><br><i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i> <ul style="list-style-type: none"><li>Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.</li></ul>                                                                                                                                                              | <i>May '24 – Jul '25</i> |
| <b>Volunteer</b><br><i>Junoon, National Service Scheme (NSS), BITS Pilani</i> <ul style="list-style-type: none"><li>Organized sports and recreational competitions for specially abled children from 25+ colleges.</li></ul>                                                                                                                                                                                 | <i>Sep '23 – Sep '23</i> |

## TEACHING

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| <b>Teaching Assistant</b><br><i>CIS 4190/5190: Applied Machine Learning, University of Pennsylvania</i>                                                                                                                                                                                                                          | <i>Aug '26 – Dec '26</i> |
| <i>CS F320: Foundations of Data Science, BITS Pilani</i>                                                                                                                                                                                                                                                                         | <i>Jan '26 – May '26</i> |
| <i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i>                                                                                                                                                                                                                                                                        | <i>Jan '25 – May '25</i> |
| <i>ME F216: Materials Science and Engineering, BITS Pilani</i> <ul style="list-style-type: none"><li>Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications.</li><li>Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.</li></ul> | <i>Sep '24 – Dec '24</i> |

## INDUSTRY EXPERIENCE

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| <b>ML Intern   Uber (AI Solutions)</b><br><i>Supervisor: <a href="#">Mr. Siddharth Malreddy</a>, Tech Lead Manager &amp; <a href="#">Mr. Ishan Nigam</a>, Senior ML Engineer, Uber</i>                                                                                                                              | <i>Jul '25 – Dec '25</i> |
| <ul style="list-style-type: none"><li>Streamlined object tracking in <a href="#">uLabel</a> for RGB/IR videos, flagging object annotation anomalies via XGBoost (mAP50: 75%).</li><li>Screened object-association errors with a <a href="#">TrackTrack</a>-based cross-frame consistency check (F1: 88%).</li></ul> |                          |
| <b>Subject Matter Expert   Knowledge Cultivator</b><br><i>Self-supervised</i>                                                                                                                                                                                                                                       | <i>Nov '23 – Feb '24</i> |
| <ul style="list-style-type: none"><li>Resolved numerical problems for student clients using MATLAB, CAD, and Python.</li><li>Authored academic content across diverse research topics.</li></ul>                                                                                                                    |                          |

AWARDS & ACHIEVEMENTS

|                                                                                                |         |
|------------------------------------------------------------------------------------------------|---------|
| Finalist — Meta PyTorch OpenEnv Hackathon (India) <a href="#">🔗</a>                            | Apr '26 |
| 3 <sup>rd</sup> Place — MathWorks Global Drone Student Challenge 2025                          | Mar '25 |
| Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) <a href="#">🔗</a>   | Jul '24 |
| Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 <a href="#">🔗</a>    | Jul '24 |
| 5 <sup>th</sup> Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals) | Sep '23 |

PROFESSIONAL ACTIVITIES

|                                |      |
|--------------------------------|------|
| Graduate Student Member — IEEE | 2026 |
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COMPETITIVE EXAMINATIONS

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|------------------------------------------------------------------------|------|
| GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)                      | 2025 |
| TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28) | 2025 |
| JEE Advanced: All India Rank 14372                                     | 2022 |

SKILLS

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| Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing |
| Programming Languages: Python, C++, C, C#, Java, Bash (Shell)                                                                                                                       |
| Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace                                                                                                 |